

LAB07

Danillo dos Santos Lemes

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.6.1

```
conn <- dbConnect(SQLite(),"uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
SELECT DISTINCT i.name, i.id FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
ORDER BY i.name;
```

Table 1: Displaying records 1 - 10

name	id
ABIGAIL BENZINE	4787911
ADIN-CRISTIAN ANDREI	3111842
AKICHIKA OZEKI	3775809
ALAN HUANG	4958796
ALBERTO DEL PIA	5451706
ALBRECHT KLEMM	3661782

name	id
ALEXANDER COVINGTON	5656034
ALEXANDER FISH	4680082
ALEXANDER KISELEV	3573775
ALYSSA DIGILIO	5162339

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
SELECT i.name AS professor_name, SUM(ga.a_count * 4.0 + ga.ab_count * 3.5 + ga.b_count * 3.0) AS gpa_medio
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions ga ON sec.course_offering_uuid = ga.course_offering_uuid AND sec.nu
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio
LIMIT 1
```

Table 2: 1 records

professor_name	gpa_medio
JAMES D KUELBS	2.6375

- Quem é o professor mais fácil?

```
SELECT i.name AS professor_name, SUM(ga.a_count * 4.0 + ga.ab_count * 3.5 + ga.b_count * 3.0) AS gpa_medio
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions ga ON sec.course_offering_uuid = ga.course_offering_uuid AND sec.nu
WHERE s.abbreviation = 'STAT'
```

```
GROUP BY i.id, i.name
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio DESC
LIMIT 1
```

Table 3: 1 records

professor_name	gpa_medio
SRIKANTHMADHAVAN ARAVAMUTHAN	4

- Qual é a disciplina mais difícil?

```
SELECT c.name AS course_name, c.number AS course_number, SUM(ga.a_count * 4.0 + ga.ab_count * 3.0) AS gpa_medio
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions ga ON co.uuid = ga.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.name, c.number
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio
LIMIT 1
```

Table 4: 1 records

course_name	course_number	gpa_medio
Introduction to the Theory of Probability	431	2.886374

- Qual é a disciplina mais fácil?

```
SELECT c.name AS course_name, c.number AS course_number, SUM(ga.a_count * 4.0 + ga.ab_count * 3.0) AS gpa_medio
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions ga ON co.uuid = ga.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.name, c.number
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio DESC
LIMIT 1
```

Table 5: 1 records

course_name	course_number	gpa_medio
Data Science Practicum	628	4

4. Desconecte do banco de dados

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-24 00:13:18 -03"
```